- eigenvectors and eigenvalues
- How to find e.v. ?

§5.1 Eigenvectors and eigenvalues (square matrices) (Part I)

Matrix multiplication can have quite different effects on different vectors

Exmp 1 $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$, $\vec{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$A\vec{u} = \begin{bmatrix} 3\times(-1) + (-2)\times 1 \\ 1\times(-1) + 0\times 1 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$$

$$A\vec{v} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2\vec{v}.$$

Def A : $n \times n$ matrix.

An eigenvector of A is a nonzero vector $\vec{x} \in \mathbb{R}^n$ s.t. $A\vec{x} = \lambda\vec{x}$ for some scalar λ.

This scalar λ is an eigenvalue of A.

This $\vec{x}$ is an eigenvector corresponding to λ.

Exmp $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ $\vec{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

Are $\vec{u}$ & $\vec{v}$ eigenvectors?

$$A\vec{u} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4\begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4\vec{u}$$

$$A\vec{v} = \begin{bmatrix} -9 \\ 11 \end{bmatrix}$$ not a multiple of $\vec{v}$

Thus $\vec{u}$ is an eigenvector corresponding to (-4), and $\vec{v}$ is not an eigenvector.

How to look for (eigenvalues and) eigenvectors?

Exmp3 Show that 7 is an eigenvalue of matrix, and find the corresponding eigenvect.

Sol The scalar 7 is an eigenvalue of A iff.

$$\boxed{A\vec{x} = 7\vec{x}}$$

has a nontrivial sol. It's equivalent to

$$\boxed{(A - 7I)\vec{x} = 0}$$ (homogeneous equation)

Form the matrix

$$A - 7I = \begin{bmatrix} 1-7 & 6 \\ 5 & 2-7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

The equation has a nontrivial sol. Thus 7 is an eigenvalue.

Eigenvectors

Augmented matrix $\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

The general solution has the form $x_2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $x_2 \in \mathbb{R}$.

Each vector $x_2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $x_2 \in \mathbb{R}$, is an eigenvector corresponding to $\lambda = 7$.

Def The set of all solutions of $(A - \lambda I)\vec{x} = \vec{0}$ is the eigenspace of A corresponding to the eigenvalue λ.

Rmk
- It's a subspace of $\mathbb{R}^n$
- It consists of $\vec{0}$ and all the eigenvectors corresponding to λ.

Exmp. 4 $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ An eigenvalue = 2.

Find a basis for the corresponding eigenspace.

Sol (solution set of $(A-2I)\vec{x} = \vec{0}$ i.e. null space of $A-2I$.)

$$A - 2I = \begin{bmatrix} 4-2 & -1 & 6 \\ 2 & 1-2 & 6 \\ 2 & -1 & 8-2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

Augmented matrix of the equation $(A-2I)\vec{x} = \vec{0}$:

$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

It has nontrivial solutions.

2 → indeed an eigenval.

general solution : $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 - 3x_3 \\ x_2 \\ x_3 \end{bmatrix}$

$$= x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \qquad x_2, x_3 \in \mathbb{R}$$

a basis for the eigenspace : $\left\{ \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ ☐

Thm 2. $\vec{v}_1, \cdots, \vec{v}_r$ eigenvectors corr to distinct eigenvalues $\lambda_1, \cdots, \lambda_r$ of an $n \times n$ matrix A. Then the set $\{\vec{v}_1, \cdots, \vec{v}_r\}$ is linearly independent.

Pf Exmp $\vec{v}_1, \vec{v}_2$ eigenvector corr to two distinct eigenvalues

Suppose $\exists$ scalars a_1 & a_2 λ_1 & λ_2

s.t. $a_1\vec{v}_1 + a_2\vec{v}_2 = \vec{0}$

Then $A(a_1\vec{v}_1 + a_2\vec{v}_2) = a_1\lambda_1\vec{v}_1 + a_2\lambda_2\vec{v}_2 = \vec{0}$

and $\lambda_1(a_1\vec{v}_1 + a_2\vec{v}_2) = \vec{0}$

$\lambda_2(a_1\vec{v}_1 + a_2\vec{v}_2) = \vec{0}$

We can see that $\lambda_1 a_2 \vec{v}_2 = \lambda_2 a_2 \vec{v}_2$

$\lambda_2 a_1 \vec{v}_1 = \lambda_1 a_1 \vec{v}_1$

Since $\lambda_1 \neq \lambda_2$, $a_2\vec{v}_2 = \vec{0} = a_1\vec{v}_1$

Thus $a_1 = a_2 = 0$

Therefore $\vec{v}_1$ & $\vec{v}_2$ are linearly independent

Eigenvectors and difference equations

A = $n \times n$ matrix.

$$\vec{x}_{k+1} = A\vec{x}_k \quad (k = 0, 1, 2, \cdots)$$

Take an eigenvector $\vec{x}_0$ corr. to the eigenvalue λ and let

$$x_k = \lambda^k \vec{x}_0 \quad (k = 1, 2, \cdots) \qquad (9)$$

This is a sol. to the difference equation:

$$A\vec{x}_k = A(\lambda^k \vec{x}_0) = \lambda^k (A\vec{x}_0) = \lambda^{k+1}\vec{x}_0 = \vec{x}_{k+1}.$$

Linear comb. of sol. in the form of (9) are sol. too.